

On device: cuda
Loading Jena Climate dataset from local file: jena_climate_2009_2016.csv...
Generating sequences (72-hour input, 12-hour forecast)...

--- Training Transformer ---

Epoch 01	Train: 0.08516	Val: 0.03108
Epoch 02	Train: 0.03395	Val: 0.02786
Epoch 03	Train: 0.02942	Val: 0.02332
Epoch 04	Train: 0.02732	Val: 0.02468
Epoch 05	Train: 0.02637	Val: 0.02439
Epoch 06	Train: 0.02540	Val: 0.02447
Epoch 07	Train: 0.02407	Val: 0.02134
Epoch 08	Train: 0.02352	Val: 0.02122
Epoch 09	Train: 0.02331	Val: 0.02169
Epoch 10	Train: 0.02297	Val: 0.02073
Epoch 11	Train: 0.02283	Val: 0.02206
Epoch 12	Train: 0.02278	Val: 0.02266
Epoch 13	Train: 0.02257	Val: 0.02137
Epoch 14	Train: 0.02186	Val: 0.02128
Epoch 15	Train: 0.02168	Val: 0.02162
Epoch 16	Train: 0.02177	Val: 0.02114
Epoch 17	Train: 0.02134	Val: 0.02101
Epoch 18	Train: 0.02139	Val: 0.02068
Epoch 19	Train: 0.02120	Val: 0.02122
Epoch 20	Train: 0.02122	Val: 0.02194

Finished Transformer Training in 260.14s | Best Val Loss: 0.02068

--- Training CustomLSTM ---

Epoch 01	Train: 0.07782	Val: 0.02758
Epoch 02	Train: 0.03213	Val: 0.02473
Epoch 03	Train: 0.02889	Val: 0.02448
Epoch 04	Train: 0.02724	Val: 0.02430
Epoch 05	Train: 0.02677	Val: 0.02374
Epoch 06	Train: 0.02604	Val: 0.02346
Epoch 07	Train: 0.02561	Val: 0.02231
Epoch 08	Train: 0.02529	Val: 0.02302
Epoch 09	Train: 0.02496	Val: 0.02269
Epoch 10	Train: 0.02484	Val: 0.02220
Epoch 11	Train: 0.02453	Val: 0.02339
Epoch 12	Train: 0.02454	Val: 0.02223
Epoch 13	Train: 0.02430	Val: 0.02215

Epoch 14		Train: 0.02412		Val: 0.02205
Epoch 15		Train: 0.02417		Val: 0.02160
Epoch 16		Train: 0.02402		Val: 0.02269
Epoch 17		Train: 0.02389		Val: 0.02175
Epoch 18		Train: 0.02356		Val: 0.02276
Epoch 19		Train: 0.02297		Val: 0.02212
Epoch 20		Train: 0.02287		Val: 0.02196

Finished CustomLSTM Training in 601.58s | Best Val Loss: 0.02160

--- Comparative Evaluation Metrics (in Celsius) ---

Transformer (Primary):

MSE: 2.7898
MAE: 1.2338
Runtime: 260.14s

Custom LSTM (Comparison):

MSE: 2.9241
MAE: 1.2566
Runtime: 601.58s